Fiscal Year:	FY 2019	**Task Last Updated:**	FY 09/21/2018
PI Name:	Su, Ching-Hua Ph.D.		
Project Title:	Crystal Growth of II-VI Compound Semiconductors		
Division Name:	Physical Sciences		
Program/Discipline:	MATERIALS SCIENCE		
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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PI Organization Type:	NASA CENTER	**Phone:**	256-544-7776
Organization Name:	NASA Marshall Space Flight Center		
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PI Web Page:			
City:	Huntsville	**State:**	AL
Zip Code:	35812	**Congressional District:**	5
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	NOT AVAILABLE
Start Date:	10/01/2006	**End Date:**	09/30/2021
No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA MSFC
Contact Monitor:	Rogers, Jan	**Contact Phone:**	256.544.1081
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Flight Program:	ISS		
Flight Assignment:	NOTE: Changed end date to 9/30/2021 per L. Strutzenberg/MSFC (Ed., 7/23/19) NOTE: Changed end date to 10/01/2019, per PI (Ed., 9/21/18) NOTE: Changed end date to 09/30/2018, per PI (Ed., 10/18/16); note revised to 10/1/2018 to accommodate final reporting (Ed., 1/19/18) NOTE: Change in end date per P. Downey/MSFC (Ed., 9/28/2011)		
Key Personnel Changes/Previous PI:	FY2018 report: Prof. N. B. Singh, University of Maryland, Baltimore County and Prof. M. Dudley, Stony Brook University are supporting this project. Previous Co-Investigator Dr. Narayanan Ramachandran is no longer CoI on the project.		
COI Name (Institution):			
Grant/Contract No.:	Internal Project		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Two reasons have been put forward to account for the improved crystal quality when crystals are grown in a microgravity environment. The first is weight-related reductions in crystal strain and defects. These are thought to be caused by the weight of the crystals during processing at elevated temperatures and retained on cooling, particularly for materials with low yield strength. The second and more general reason is related to the reduction in density-gradient driven convection in the fluid during crystal growth in reduced gravity. The objectives and experiment description for the proposed flight experiment “Growth of Ternary Compound Semiconductors” by (1) vapor growth and (2) melt growth are described below. 1a) Vapor Growth The physical vapor transport (PVT) crystal growth process consists of essentially three processes: sublimation of source material, transport of vapor species, and condensation of vapor species to form crystal. The latter two processes can be affected by the convection caused by gravitational accelerations on Earth. Reductions in such convection in low gravity is expected to yield a nearly diffusion-limited growth condition which results in more uniform growth rates (on the microscopic scale) and hence greater crystalline perfection and compositional homogeneity. This reduction of convective contamination in a reduced gravity environment will help to simplify the complexity of the coupled mass transport and growth kinetics problem. An improved comparison between the experimental results and theoretical simulations is also expected and will be beneficial to any growth process involving vapor mass transport or vapor-crystal interface growth kinetics. The materials to be studied in this project are ZnSe and the related ternary compound semiconductors such as ZnSeTe, ZnSeS, and ZnCdSe. The scientific objectives of the growth of ZnSe crystals in reduced gravity using physical vapor transport (PVT) processes are: • to establish the relative contributions of gravity-driven fluid flows to (i) the non-uniform incorporation of impurities and defects and (ii) the deviation from stoichiometry observed in the grown crystals as a result of buoyancy-driven convection and irregular fluid-flows caused by growth interface fluctuations and interface evolution. • to assess self-induced strain effects developed during processing at elevated temperatures and retained on cooling caused by the weight of the crystals from x-ray topography and by measuring the density and distribution of various crystalline defects. • to obtain high quality space-grown materials for various thermophysical and electrical properties measurements and as substrates for device fabrication and thus assess device performance as influenced by a substantial reduction in gravity-related effects. 1b) Melt Growth The melt growth of homogeneous ternary compound semiconductors, such as cadmium zinc telluride (Cd1-xZnxTe, $0 < x < 1$) is difficult because the thermal and compositional gradients in the melt just ahead of the growth interface during directional solidification can cause substantial gravity-driven fluid flows that alter the relative distribution of the two (CdTe and ZnTe) binary components and introduce structural defects into the grown crystals. The scientific objectives of the growth of CdZnTe crystals in reduced gravity using directional solidification process are: • to investigate quantitatively the influences of gravitationally dependent phenomena on the growth process and the incorporation of impurities and defects in the resulting crystals. • to better understand and control the growth parameters in the production process on Earth so as to increase the yield of usable material from the grown ingot. • to assess strain effects developed during processing at elevated temperatures and retained on cooling, caused by the weight of the crystals and, possibly, the contact with the growth container, from the structural characterization and by measuring the density and distribution of various crystalline defects. • to obtain high quality space-grown materials for various electrical and optical property measurements and as materials for the x-ray and gamma-ray detector fabrication as well as substrates for infrared device application and thus assess device performance as influenced by a substantial reduction in gravity-related effects.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	• In crystal growth there is a need to understand the relation between processes in the fluid phase (both liquid and vapor), such as buoyancy driven convection and the incorporation of impurities, and defects in the resulting crystal • Relation between fluid phase processes and the generation of defects in a grown crystal is an outstanding problem in materials growth • Crystal quality greatly influences important electronic properties in materials • Ternary compound semiconductors are of vital national interest as sensors in x-ray telescopes and for homeland security, and as substrate materials for infrared sensors • Studies in microgravity will be compared with modeling and will be used to optimize ground-based experiments
Task Progress:	The main disadvantages of the existing vapor growth techniques, such as physical vapor transport, are that (a) the significant amount of waste from starting material, and (b) the crystal growth rates are low and inconsistent which lowers the single crystal yield. In this period, progress has been made that with a different arrangement of the growth furnace, a much higher production yield can be consistently achieved and a routine production of high-quality single crystals of these semiconductors can be realized. The invention has been filed for a patent application through the Office of Chief Patent Counsel of Marshall Space Flight Center (MSFC). For the upcoming flight experiments, the integration of the crystal growth ampoule and the Sample Cartridge Assembly (SCA) has been initiated. The flight project has been officially named as “Growth of Ternary Compound Semiconductors (GTCS).” A bi-weekly meeting between Ching-Hua Su and a team of engineers from MFSC has been held and an Integration Design Review (IDR) has been ongoing from the kick-off meeting on 8/23/2018 to the Pre-Board on 9/7/2018.

Bibliography Type:	Description: (Last Updated: 02/18/2025)
Abstracts for Journals and Proceedings	Cooper C, Su CH, Choa F-S, Arnold B, Singh NB. "Design of Multifunctional Materials: Chalcogenides and Chalcopyrites." Emerging Multifunctional Materials for Bio, EO, RF and Radiation Sensors Symposium, Materials Science & Technology 2017, Pittsburgh, PA, October 8-12, 2017. Materials Science & Technology 2017, Pittsburgh, PA, October 8-12, 2017. , Oct-2017
Articles in Peer-reviewed Journals	Singh NB, Su C-H, Arnold B, Choa F-S, Cooper C. "Multifunctional 2D- materials: Gallium selenide." Materials Today:Proceedings. 2017;4(4 Pt E):5471-7. https://doi.org/10.1016/j.matpr.2017.06.002 (Special issue on International Conference on Multifunctional Materials for Device Applications (ICMDA-2016), Panta, India, October 26-28, 2016) , Jan-2017
Articles in Peer-reviewed Journals	Singh NB, Cooper C, Strobbia P, Prasad NS, Su CH, Arnold B, Choa F-S. "Nanomorphology and performance of pure and doped lead selenide for infrared detector." Optical Engineering. 2017 Jul;56(7):077106. https://doi.org/10.1117/1.OE.56.7.077106 , Jul-2017
Articles in Peer-reviewed Journals	Baird JK, Su C-H. "Acoustic vibration effects in classical nucleation theory." Journal of Crystal Growth. 2018 Apr 1;487:65-71. https://doi.org/10.1016/j.jcrysgro.2018.02.008 , Apr-2018
Articles in Peer-reviewed Journals	Su C-H. "Experimental determination of lattice thermal conductivity and Lorenz number as functions of temperature for n-type PbTe." Materials Today Physics. 2018 Jun;5:58-63. https://doi.org/10.1016/j.mtphys.2018.05.005 , Jun-2018
Articles in Peer-reviewed Journals	Su C-H. "Processing high efficiency functional gradient thermoelectric materials." Materials Today Physics. 2018 Aug;6:1-8. https://doi.org/10.1016/j.mtphys.2018.06.001 , Aug-2018
Papers from Meeting Proceedings	Singh NB, Su C-H, Arnold B, Cullum B, Choa F-S, Carpenter T, Sachs D, Mandal KD. "Growth of bio sensor materials by physical vapor transport method." SPIE Commercial + Scientific Sensing and Imaging 2018, Orlando, Florida, April 15-19, 2018. In: Smart Biomedical and Physiological Sensor Technology XV Proceedings, Vol 10662: 1066203. https://doi.org/10.1117/12.2303665 , Apr-2018
Patents	Serial No.: 15/983,847. Prepared by legal service firm – Precipice IP, PLLC; Filed by the Office of Chief Patent Counsel of MSFC: 5/18/2018. May-2018 Su C-H. "A Process for Producing Crystals by Physical Vapor Transport."